

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Case Study

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO3 – Precision	Assessment:	Assessment Tool 2 – Case Study
Weightage:	40% of CIA		
CO5 Statement:	Apply N-gram models, smoothing techniques, part-of-speech tagging systems and to evaluate effective syntactic analysis. (BL-3)		
Student Name:	SYED MUHAFIZ A	Reg. No.:	192472282
Section / Batch:		Date:	

Code of Conduct

I SYED MUHAFIZ A (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Instructions

- Read all three case studies carefully before attempting the questions.
- Provide appropriate justifications and interpretations for every computed result.
- Use NLP concepts, algorithms, and terminology wherever applicable.
- Diagrams, flowcharts, tables, or mathematical formulations may be used to support your answers.
- Duration: 30 minutes.

Section / Topic	Focus	Case Study
N-Gram Language Models and Smoothing Techniques	Bigram and Trigram probability computation, Maximum Likelihood Estimation (MLE), Backoff models, Deleted Interpolation, Entropy calculation, uncertainty analysis, real-time next-word prediction	Case Study 1 – Smart Mobile Keyboard Prediction System
Part-of-Speech (POS) Tagging and Word Classes	Penn Treebank tagset, contextual ambiguity resolution, rule-based tagging, stochastic tagging (HMM), emission and transition probabilities, word class identification, chatbot language understanding	Case Study 2 – AI-Powered Customer Support Chatbot
Transformation-Based Tagging (Brill Tagger)	POS tag correction, transformation rules, contextual tagging, linguistic validation, tag sequence refinement, ambiguity handling	Case Study 3 – News Analytics and POS Tag Correction System
Corpus Statistics and Probabilistic Analysis	Word frequency counting, corpus-based probability estimation, entropy-based uncertainty measurement, tagging confidence evaluation	Case Study 3 – News Analytics and POS Tag Correction System
Integrated Syntactic Analysis and NLP Applications	Application of language models, POS tagging, probabilistic methods, corpus analysis, ambiguity resolution, and decision-making in real-world NLP systems	Case Studies 1, 2, and 3

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Q. No. / Criteria Level	Understanding of Case	Application of Concepts	Analysis	Solution Design	Clarity	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

CASE STUDY 1: Smart Mobile Keyboard Prediction System

1. Compute the MLE of Bigram Probability $P(\text{science} \mid \text{data})$

The Maximum Likelihood Estimate (MLE) of a bigram probability is calculated using the frequency of the bigram divided by the frequency of the previous word.

$$P(\text{science} \mid \text{data}) = C(\text{data science}) / C(\text{data})$$

Given: $C(\text{data science}) = 3$ and $C(\text{data}) = 3$.

$$P(\text{science} \mid \text{data}) = 3 / 3 = 1.0$$

Answer: $P(\text{science} \mid \text{data}) = 1.0$ or 100%.

This means that in the given training corpus, every occurrence of “data” is followed by “science”. Therefore, the mobile keyboard can confidently predict “science” when the user types “data”. In a real predictive keyboard, a high probability means that the word is a strong candidate for next-word prediction. However, this probability is based only on the small corpus provided; a larger corpus would normally produce a less extreme probability.

2. Backoff Model for the unseen sequence “data science improves”

The sequence “data science improves” is unseen because the trigram does not occur in the corpus. An unsmoothed trigram model would assign zero probability to the unseen continuation.

$$P(\text{improves} \mid \text{data, science}) = 0$$

A Backoff Model solves this problem by moving to a lower-order model when the higher-order N-gram is unavailable:

$$P(\text{improves} \mid \text{data, science}) \rightarrow P(\text{improves} \mid \text{science}) \rightarrow P(\text{improves})$$

The supplied corpus does not contain the word “improves”, and its unigram count is not provided. Therefore, an exact numerical probability for the complete sequence cannot be calculated from the supplied statistics. However, the first bigram can be calculated as $P(\text{science} \mid \text{data}) = 1$.

Backoff is essential in real-time predictive text because users constantly produce new and uncommon word combinations. If every unseen sequence were assigned zero probability, many perfectly valid sentences would become impossible for the prediction system. Backoff allows the system to use trigram evidence when available, bigram evidence when the trigram is unavailable, and unigram evidence as a final fallback. This improves coverage and prevents zero-probability problems.

3. Deleted Interpolation for “data science is”

Deleted interpolation combines probabilities from trigram, bigram, and unigram language models. The formula is:

$$P = \lambda_1 P(\text{trigram}) + \lambda_2 P(\text{bigram}) + \lambda_3 P(\text{unigram})$$

Given $\lambda_1 = 0.5$, $\lambda_2 = 0.3$, $\lambda_3 = 0.2$ and $P(\text{is} \mid \text{science}) = 0.66$.

$$P(\text{data science is}) = 0.5P(\text{is} \mid \text{data, science}) + 0.3(0.66) + 0.2P(\text{is})$$

$$P(\text{data science is}) = 0.5P(\text{is} \mid \text{data, science}) + 0.198 + 0.2P(\text{is})$$

The exact final numerical value cannot be calculated from the supplied information because the unigram probability $P(\text{is})$ and the exact trigram probability $P(\text{is} \mid \text{data, science})$ are not explicitly supplied.

The interpolation balances reliability and coverage. The trigram receives the highest weight and therefore provides the strongest contextual information. The bigram provides useful context when trigram evidence is weak, while the unigram provides general word-frequency information and coverage. This makes the model more robust when the training corpus is sparse.

4. Entropy of $P(\text{is}) = 0.66$ and $P(\text{drives}) = 0.33$

Entropy measures the uncertainty in a probability distribution.

$$H(X) = -\sum P(x) \log_2 P(x)$$

$$H = -(0.66 \log_2 0.66 + 0.33 \log_2 0.33)$$

$$H \approx 0.92 \text{ bits}$$

Answer: Entropy ≈ 0.92 bits.

The entropy is relatively low because “is” has a substantially higher probability than “drives”. Therefore, the keyboard has reasonably high confidence about the next word. Lower entropy indicates a more predictable distribution, while higher entropy indicates greater uncertainty. In an intelligent keyboard, entropy can therefore be used to estimate how confident the system is about its next-word predictions.

CASE STUDY 2: AI-Powered Customer Support Chatbot

1. Assign Penn Treebank POS Tags

Sentence 1: “Book a flight ticket now.”

Word	POS Tag	Explanation
Book	VB	Base-form verb; used as a command
a	DT	Determiner
flight	NN	Singular noun
ticket	NN	Singular noun
now	RB	Adverb

Therefore: Book/VB a/DT flight/NN ticket/NN now/RB.

Here, “Book” is a verb because the sentence is a command or instruction. The user is asking the system to perform an action.

Sentence 2: “This book is interesting.”

Word	POS Tag	Explanation
This	DT	Determiner

book	NN	Singular noun
is	VBZ	Third-person singular verb
interesting	JJ	Adjective

Therefore: This/DT book/NN is/VBZ interesting/JJ.

Here, “book” is a noun because it refers to an object and acts as the subject of the sentence. The same word therefore receives different tags depending on context. This is lexical ambiguity.

2. HMM likelihood of “book” as VB

Given $P(\text{book} \mid \text{VB}) = 0.6$ and $P(\text{Start} \rightarrow \text{VB}) = 0.5$.

$$\begin{aligned} P(\text{VB}, \text{book}) &= P(\text{Start} \rightarrow \text{VB}) \times P(\text{book} \mid \text{VB}) \\ &= 0.5 \times 0.6 = 0.30 \end{aligned}$$

For comparison, the noun interpretation is:

$$\begin{aligned} P(\text{NN}, \text{book}) &= P(\text{Start} \rightarrow \text{NN}) \times P(\text{book} \mid \text{NN}) \\ &= 0.5 \times 0.4 = 0.20 \end{aligned}$$

Since 0.30 is greater than 0.20, the HMM favors the VB interpretation. The starting probabilities are equal, so the higher emission probability $P(\text{book} \mid \text{VB}) = 0.6$ is responsible for the higher likelihood.

3. Rule-Based Tagging vs Stochastic/HMM Tagging

Rule-Based Tagging	Stochastic/HMM Tagging
Uses manually written linguistic rules	Uses probabilities learned from training data
Requires extensive rule creation	Learns patterns statistically
Can work well for known grammatical patterns	Handles lexical ambiguity using probabilities
Difficult to maintain at large scale	More scalable with sufficient training data
Less adaptive to new patterns	Can adapt to patterns found in data

For a large-scale airline chatbot, stochastic/HMM tagging is generally more suitable because the system must process many types of user messages and resolve ambiguous words using context. However, in practical systems, a hybrid approach can combine statistical methods with reliable linguistic rules. Modern systems may also use neural language models for stronger contextual understanding.

4. Role of POS Tagsets and Word Classes

A standardized POS tagset provides a consistent representation of grammatical information. This helps the chatbot understand the structure and intent of user messages.

POS information can improve intent detection, ambiguity resolution, syntactic analysis, and response generation. For example, in “Book a flight”, identifying “book” as VB helps the chatbot understand that the user is requesting an action. In “This book is interesting”, identifying “book” as NN prevents the system from interpreting it as an instruction.

Therefore, standardized POS tags provide useful linguistic features for downstream NLP tasks and can improve the overall reliability of conversational systems.

CASE STUDY 3: News Analytics and POS Tag Correction System

1. Apply the Transformation Rule

Initial tags: economic/JJ growth/NN increases/NNS employment/NN.

The learned rule is: Change NNS to VBZ if the preceding word is tagged NN.

The word “increases” is tagged NNS and its preceding word “growth” is tagged NN. Therefore, the rule applies:

increases/NNS → increases/VBZ

Corrected POS sequence:

economic/JJ growth/NN increases/VBZ employment/NN

The correction is linguistically appropriate because “increases” is the main verb of the sentence and agrees with the singular subject “growth”.

2. Analyze the Initial Tagging Errors

The initial tagging error occurs with “increases”. The sentence is “Economic growth increases employment.” “Economic” modifies “growth”, so it is an adjective (JJ). “Growth” is a singular noun (NN) and forms the subject. “Increases” is the third-person singular present-tense verb, so it should be tagged VBZ. “Employment” is a noun and functions as the object of the verb.

The grammatical structure can therefore be represented as: [JJ economic] [NN growth] [VBZ increases] [NN employment]. The transformation rule correctly uses the context of the preceding NN to change the incorrect NNS tag to VBZ.

3. Word Frequency Distribution

Given frequencies: economic = 120, growth = 450, increases = 210, employment = 380.

Total = 120 + 450 + 210 + 380 = 1160

Word	Frequency	Probability	Percentage
economic	120	0.1034	10.34%
growth	450	0.3879	38.79%
increases	210	0.1810	18.10%
employment	380	0.3276	32.76%
Total	1160	1.0000	100%

For example, $P(\text{growth}) = 450/1160 = 0.3879$. Frequency information helps a probabilistic POS tagger estimate how common words are in a corpus. However, frequency alone cannot determine the correct POS because a word may have multiple grammatical roles. Context, transition probabilities, emission probabilities, and transformation rules are also important.

4. Transformation-Based Tagging and Entropy

Transformation-Based Tagging, commonly associated with the Brill Tagger, works by first assigning initial POS tags and then applying learned contextual correction rules. In this case, the initial sequence contains increases/NNS. The learned rule recognizes that an NNS tag should be changed to VBZ when the preceding word is tagged NN.

economic/JJ growth/NN increases/NNS employment/NN

→ **economic/JJ growth/NN increases/VBZ employment/NN**

This improves accuracy because the tagger uses grammatical context instead of relying only on the initial tag. It is particularly useful when words have multiple possible grammatical categories.

Entropy can be used to measure uncertainty in tag assignment. Higher entropy means that the system is less certain about which tag is correct, while lower entropy means that one tag is strongly preferred.

$$H(X) = -\sum P(x) \log_2 P(x)$$

For illustration, if the initial system assigns $P(NNS) = 0.4$ and $P(VBZ) = 0.6$, the entropy is approximately 0.971 bits. If a reliable transformation makes the system much more confident, for example $P(VBZ) = 0.95$ and $P(NNS) = 0.05$, the entropy falls to approximately 0.286 bits.

Therefore, a successful transformation can reduce uncertainty and improve confidence in the final POS assignment. Transformation-Based Tagging improves the output by combining an initial tagging stage with context-sensitive learned rules.

Overall Conclusion

These three case studies demonstrate important NLP techniques used in practical systems. N-gram models, backoff, interpolation, and entropy help predictive keyboards estimate and manage uncertainty. POS tagging and HMMs help conversational systems resolve lexical ambiguity and understand grammatical roles. Transformation-Based Tagging improves initial POS assignments by applying learned contextual rules. Together, these methods show how statistical and rule-based NLP techniques can be combined to improve language understanding and generation.